

MPLADS SENTINEL

Revised Master Solution Flow

Official-problem-aligned architecture for AI monitoring, early warning and human decision support

Core Positioning

Sentinel should not replace the existing MPLADS/eSAKSHI operational system. It should act as a read-mostly AI intelligence layer that continuously analyzes lifecycle data, detects risk signals, explains them, and routes high-risk cases to authorized officials for verification and timely corrective action.

Design Rule

The flow must follow the actual MPLADS lifecycle: recommendation and sanction -> execution and payment -> progress monitoring -> completion and asset creation. AI checks are stage-aware, so warnings can appear before, during and after execution rather than only after a project becomes problematic.



Principle: AI flags risk. Authorized humans decide.

1. What the Official Problem Actually Requires

Problem interpretation before architecture

A. Monitor the full project lifecycle

The SIH statement explicitly names project approvals/sanctions, expenditures, cost estimates, work progress, payments, completion status and asset creation.

B. Detect multiple forms of risk

Required outputs include unusual patterns, cost overruns, duplicate works, delays, irregularities, deviations from norms and potential misuse/fraud indicators.

C. Support decisions - not automate guilt

The system is expected to generate risk-based alerts, predictive insights and dashboards so authorities can take timely corrective action.

Five architecture consequences

1 Lifecycle-aware

We should not run one generic anomaly model. Checks must differ at pre-sanction, execution and completion stages.

2 Existing-system aware

eSAKSHI already supports MPLADS work/fund processes and real-time monitoring. Sentinel should layer intelligence over it instead of recreating the transaction system.

3 Trend + case intelligence

The official statement asks for trend analysis as well as project-level anomalies. Sentinel therefore needs both aggregate pattern monitoring and case investigation.

4 Role-based delivery

Members of Parliament, State Nodal Authorities, District Authorities and the Ministry need different scopes of dashboards and alerts.

5 Human-in-the-loop

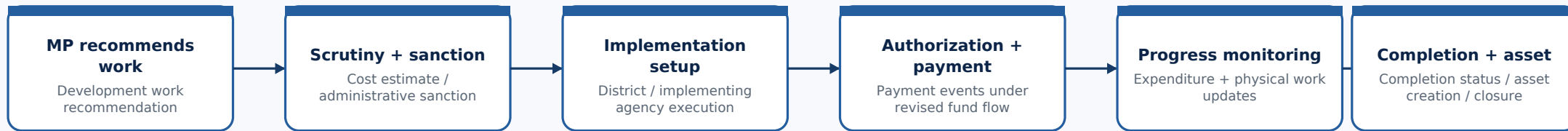
A high risk score should trigger review/verification, never an autonomous declaration of fraud, illegality or monetary loss.

Source alignment: SIH26102 problem description (MoSPI/DIID) as reproduced by SIH public mirrors; official MPLADS portal and MoSPI Annual Report 2023-24 confirm the scheme lifecycle and eSAKSHI real-time fund/work monitoring.

2. Where Sentinel Sits in the Real MPLADS Ecosystem

Architecture correction: AI sidecar, not replacement

EXISTING MPLADS / eSAKSHI OPERATIONAL LIFECYCLE (system of record)



CONTINUOUS, READ-MOSTLY DATA SYNC / EVENT FEED

MPLADS SENTINEL AI INTELLIGENCE LAYER



Why this matters

eSAKSHI already digitizes the revised MPLADS fund-flow and work process. Rebuilding it would waste hackathon effort and weaken feasibility.

Sentinel responsibility

Observe lifecycle events, derive signals, compare projects, predict risk, explain evidence, and support authorized review.

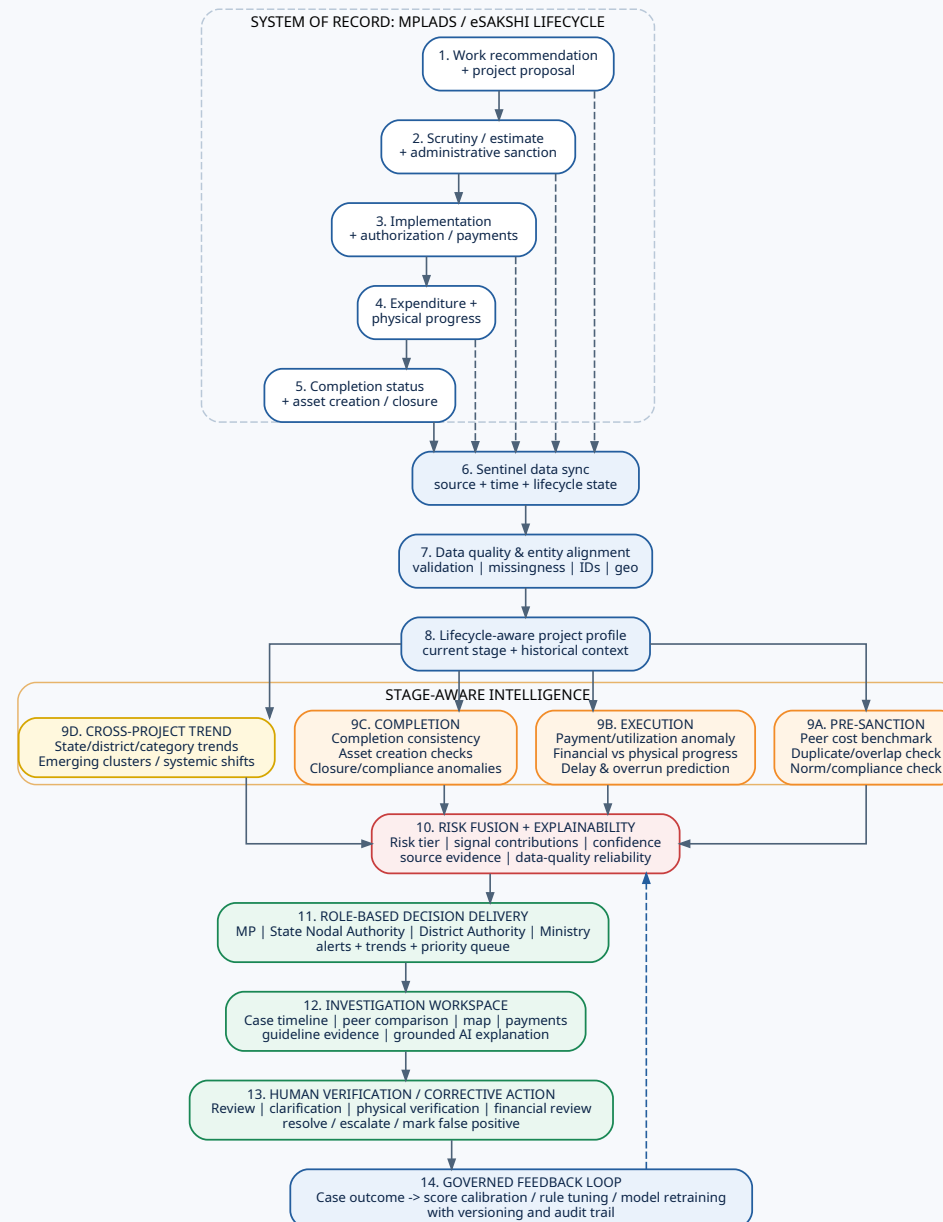
Sentinel must NOT

Autonomously approve/reject works, block payments, accuse an entity, or silently change official records. Those remain governed actions.

Official MPLADS revised fund-flow instructions state that processes from work recommendation and administrative sanction through authorization and vendor payment are handled through the web solution; payments are authorized by implementing agencies and concurred by implementing district authorities.

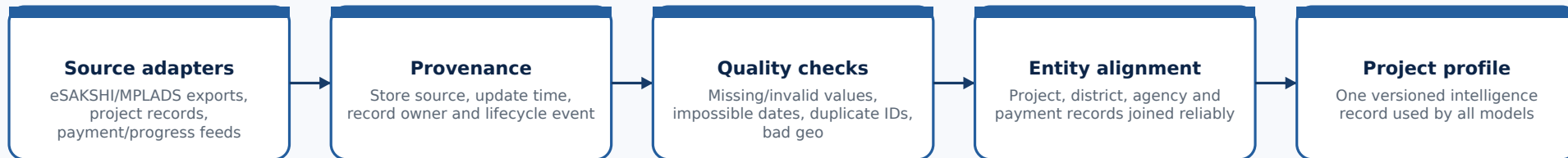
3. Revised Master Solution Flow

One end-to-end view aligned to the official problem



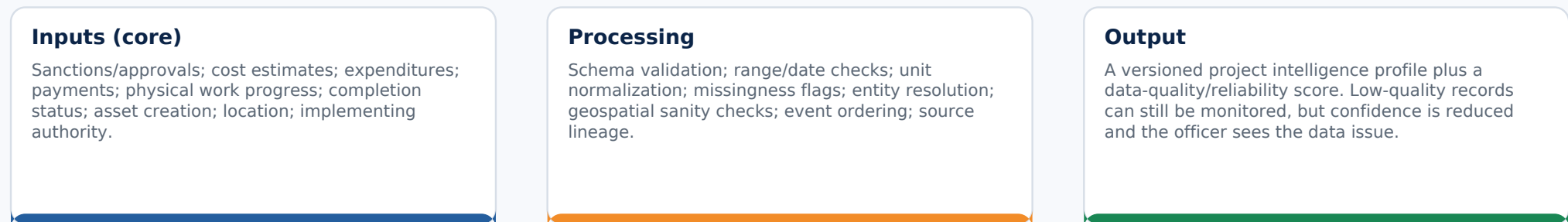
4. Step 0 - Trusted Data Foundation

Before any AI score is allowed to exist



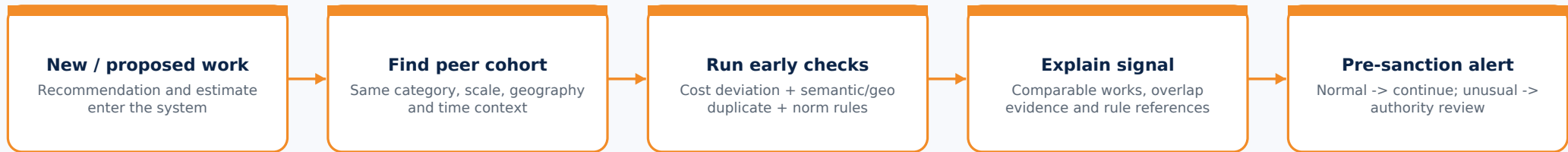
Why this stage is mandatory

Official project monitoring depends on data from many lifecycle events. If a project is missing its latest progress update, a model may incorrectly interpret old progress as a delay. If two project names refer to the same work but their IDs are inconsistent, duplicate detection becomes unreliable. Sentinel therefore carries a data-quality reliability score alongside every AI finding.

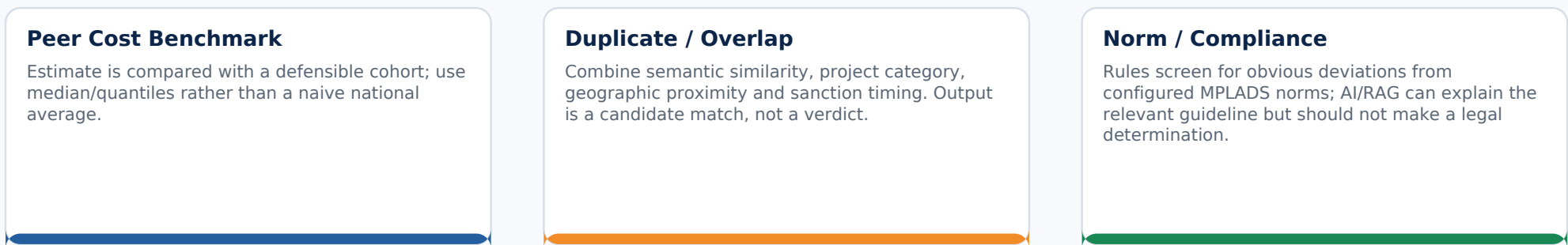


5. Stage 1 - Pre-Sanction Intelligence

Prevent avoidable risk before execution begins



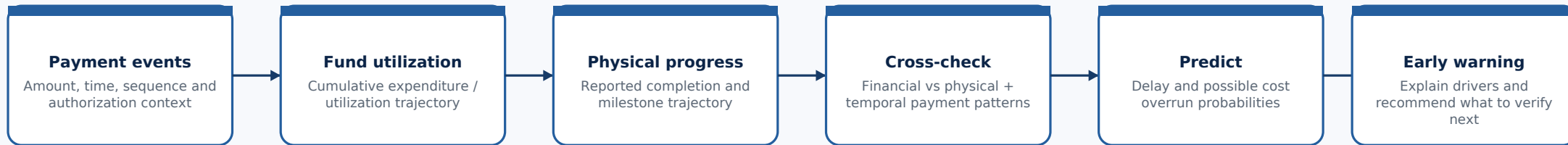
Core checks



Change from the earlier plan duplicate and cost checks move earlier, because they can create an early warning before money is substantially spent.

6. Stage 2 - Execution Intelligence

The highest-value early-warning stage



Financial-progress mismatch engine

Compare cumulative fund utilization with physical progress while accounting for project type and expected payment schedule. A large gap is a risk signal, not proof of wrongdoing. Example logic: high utilization + low physical progress + weak milestone evidence -> high verification priority.

Payment-pattern + prediction engine

Use time-series/outlier methods for sudden spending concentration, unusual payment timing or repeated patterns. Separately model delay/cost-overrun probability using elapsed time, progress velocity, historical category behavior, agency context and payment trajectory.

SIGNAL

"Financial utilization is materially ahead of reported physical progress."

ACTION

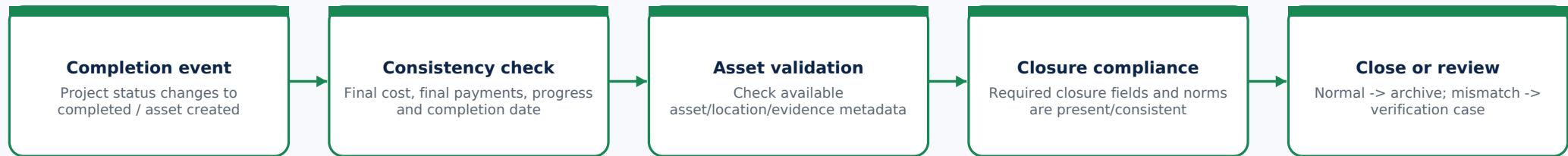
Validate progress evidence; review recent invoices/payments; consider next authorization only through authorized workflow.

PREDICT

"Significant delay probability: 81%; estimated additional delay: 4-6 months."

7. Stage 3 - Completion and Asset Intelligence

Do not stop analysis when a project is marked complete



Why this was missing in the old master flow

The official problem explicitly includes completion status and asset creation. A project that looks normal during execution can still have inconsistent final cost, late payments after completion, incomplete closure evidence or asset metadata that requires review. Completion intelligence therefore deserves its own stage.

Completion consistency

Final physical progress should align with completion status; final payment/expenditure should be coherent with sanctioned/revised cost and timeline.

Asset creation checks

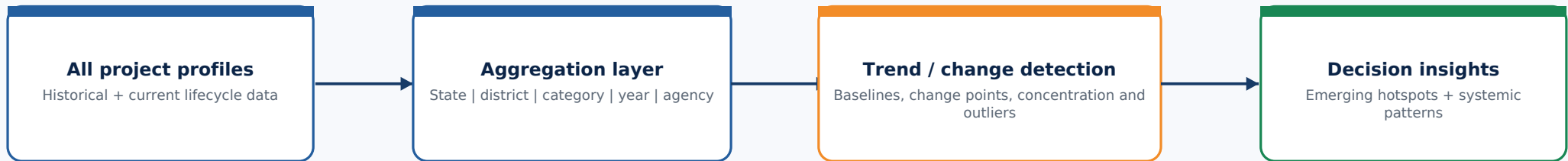
When source data provides coordinates, asset IDs, photos or handover metadata, Sentinel checks presence, duplicates and location consistency. Image verification is optional, not assumed.

Closure decision

A completed project can be archived as low risk, kept under monitoring, or routed for human verification with the exact inconsistency and source evidence shown.

8. Cross-Project Trend Intelligence

Official requirement: trends, not only individual anomalies



Examples of useful trend questions

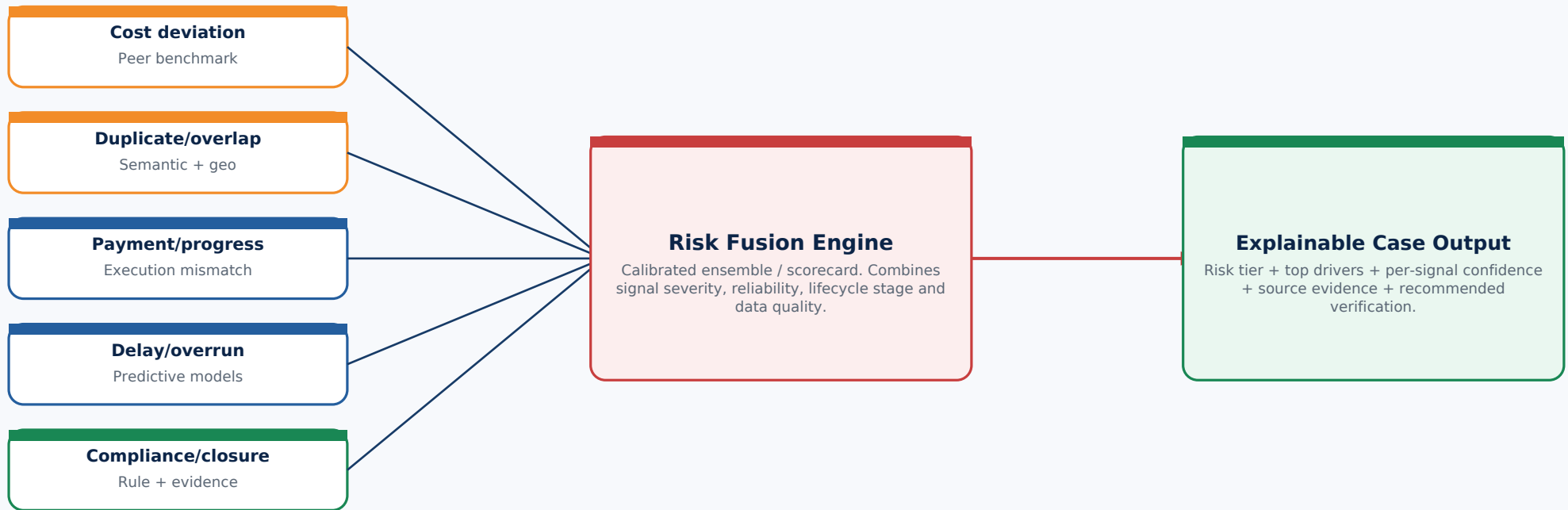
- 1 Trend question**
Which districts show a new rise in cost overruns compared with their own historical baseline?
- 2 Trend question**
Which project categories are accumulating delays this quarter?
- 3 Trend question**
Are payment concentrations or fund-utilization gaps emerging in a specific region?
- 4 Trend question**
Which implementing authorities have a statistically unusual share of high-risk works, after adjusting for volume?
- 5 Trend question**
Are duplicate/overlap signals clustering geographically?

Important design choice

Trend analytics feeds both the national command view and the case risk engine. A project can be notable because of its own values or because it sits inside an emerging systemic pattern.

9. Risk Fusion, Confidence and Explainability

Convert many signals into a defensible priority decision



1. Risk priority

How urgently should an authorized official look at this case? Example tiers: Low, Moderate, High, Critical.

2. Confidence

How reliable is each finding given model certainty, peer sample size, data quality and evidence completeness?

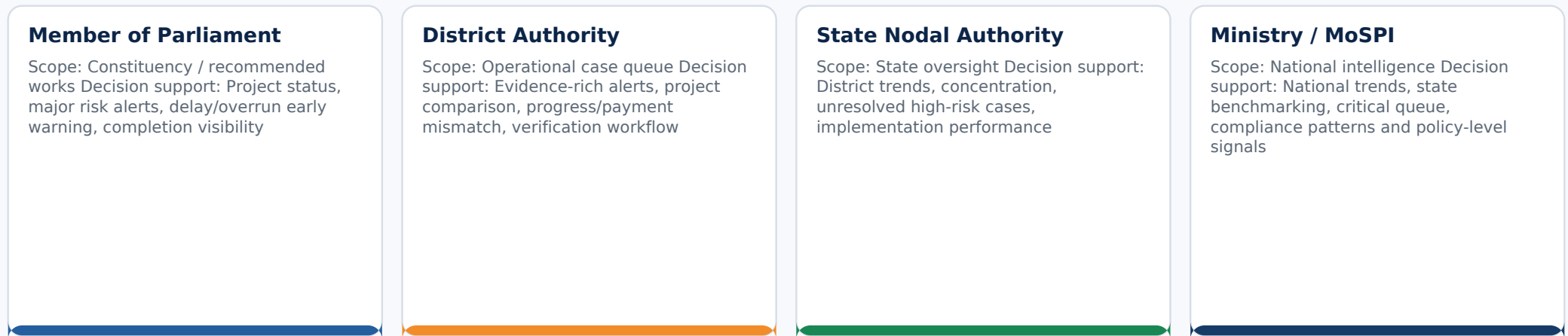
3. Evidence trail

What source records, comparable projects, timeline events and guideline references caused the alert? This makes the score contestable and auditable.

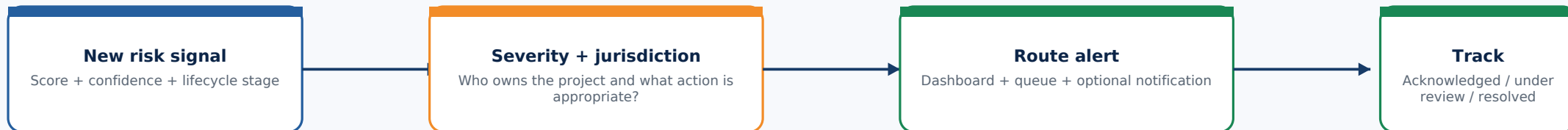
Optional derived metric: "Financial Exposure Requiring Verification" may be shown only when a defensible calculation exists; never label it as money lost.

10. Role-Based Alerts and Decision Support

Same intelligence, different scope for each authority



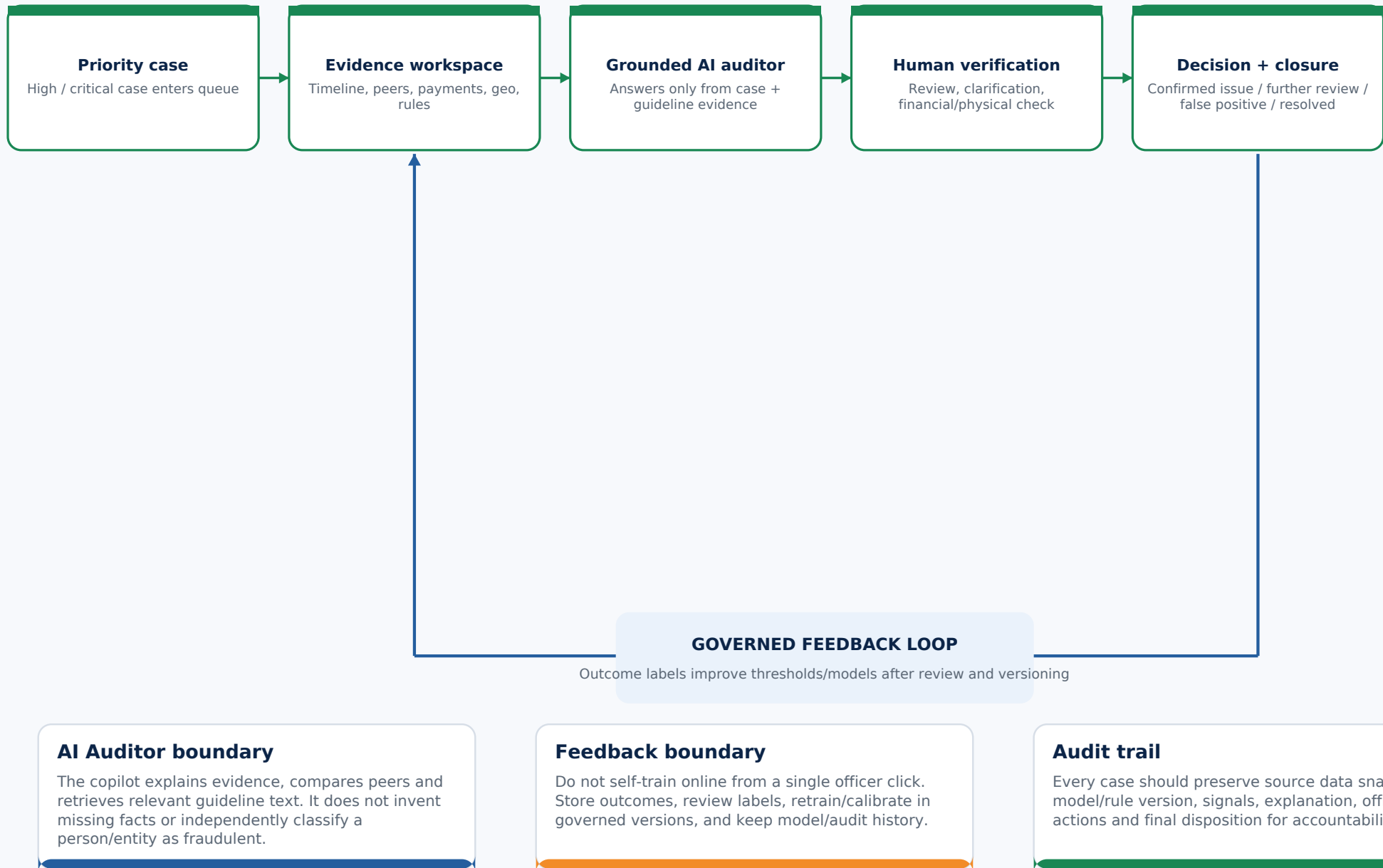
Alert routing logic



Governance rule: Sentinel may recommend review; only authorized actors take administrative, financial or enforcement action.

11. Investigation -> Verification -> Governed Learning

The final decision remains human and auditable



12. Final Flow to Use Going Forward

This replaces the earlier generic master flow



What changed from the previous master flow:

eSAKSHI is the system of record; lifecycle-aware AI replaces a flat model list; trend intelligence and asset completion are explicit; stakeholder routing is explicit; contractor/network analytics becomes optional enrichment; AI never autonomously blocks/accuses; feedback is governed.

13. Hackathon Build Priority from This Flow

What must be real in the MVP versus what can remain extension-ready

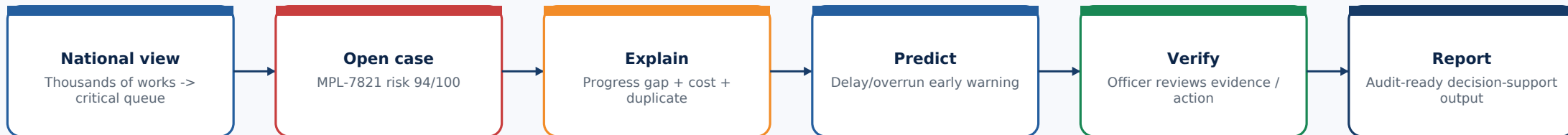
MVP - must demonstrate end-to-end

1. Ingest a synthetic/available MPLADS dataset with lifecycle fields. 2. Data quality + project profile. 3. Peer cost anomaly. 4. Duplicate/geo candidate detection. 5. Financial-vs-physical mismatch. 6. Delay prediction. 7. Risk fusion + explainability. 8. One role-based investigation queue and case page. 9. Human disposition + generated evidence report.

Extension - only if data/time supports it

1. Contractor/vendor relationship graph. 2. Advanced payment-network anomaly detection. 3. Asset image/computer-vision verification. 4. Full natural-language compliance copilot. 5. Multi-role notification workflows. 6. Model retraining dashboard. 7. External data enrichment.

Judge-ready demonstration spine



Winning discipline did one complete evidence-to-decision story before adding extra pages or AI buzzwords.

Official / source-backed facts used

- 1 SIH26102 - MoSPI / DIID problem statement**
Requires AI/ML analytics over sanctions/approvals, expenditures, cost estimates, work progress, payments, completion and asset creation; risk alerts, predictive insights, compliance, trend analysis and dashboards for MPs, State Nodal Authorities, District Authorities and the Ministry.
- 2 MPLADS official portal**
MPLADS supports MP-recommended durable community assets/basic civic amenities and the Ministry publishes guidelines, reports and monitoring material.
- 3 MoSPI Annual Report 2023-24**
Revised guidelines took effect from 1 April 2023; eSAKSHI supports the revised fund-flow procedure and real-time monitoring of funds and works by stakeholders.
- 4 Official revised fund-flow instructions**
The web solution covers recommendation, sanction, authorization and vendor payment; payment authorization/concurrence remains with designated authorities.

Proposed design choices (our architecture, not official mandates)

Lifecycle-specific AI engines; risk fusion with confidence; evidence-first explainability; read-mostly sidecar architecture; financial exposure as an optional derived metric; grounded AI auditor; governed feedback/retraining; optional graph/network and computer-vision enrichment.

<https://www.mplads.gov.in/>

<https://www.mplads.gov.in/mplads/En/2010-mplads-guidelines.aspx>

https://www.mplads.gov.in/MPLADS/UploadedFiles/Important_instructions_to_be%20followed_by_States.pdf

https://www.mospi.gov.in/sites/default/files/publication_reports/AnnualReport_2023-24.pdf

Public SIH mirrors reproducing PS SIH26102: sih2026.vuce.in / sih-fit.vercel.app (official SIH page was not machine-fetchable during research).